Seminar 3. Exploratory Data Analysis
Case: A Trade and Investment Mission to Belgium (continued)

The situation

The analytical unit has sent the head of the delegation the numbers from seminar 2. The reply came back the same evening:

> Thank you. Numbers are fine for me, but the executives at the table will not read a table. Give me five charts I can put on screen, and under each one a sentence that says what it shows.

Your task is to draw these five charts from the same dataset of Belgian production units. Each chart answers one question the delegation will face in the room.

The dataset

`Labour.csv`, 569 Belgian production units, cross-section for 1996. It is the same file as in seminar 2.

Column	Meaning	Unit
`capital`	total fixed assets at the end of 1995	millions of euro
`labour`	number of workers	persons
`output`	value added	millions of euro
`wage`	wage costs per worker	thousands of euro

Source of the data: https://rdrr.io/cran/Ecdat/man/Labour.html. Original reference: Verbeek M. (2004) A Guide to Modern Econometrics, John Wiley and Sons, chapter 4.

Before you start

1. Open JupyterLab and create a new notebook (File - New - Notebook).
2. Put `Labour.csv` next to your Jupyter notebook, in the same folder. Otherwise `read_csv` will not find the file.
3. Run a cell with **Shift + Enter**. Run the cells in order, from top to bottom: a cell can use only what the cells above it have already created.
4. Copy the cell below with **Ctrl + C** and paste it into your notebook with **Ctrl + V** (on a Mac: Cmd + C and Cmd + V). Then run it. The cell loads the data and creates a new column `size_band` that puts every firm into one of three size bands by the number of workers: Small (fewer than 50 workers), Medium (50 to 249 workers) and Large (250 workers or more). These are the thresholds of the European Commission: https://single-market-economy.ec.europa.eu/smes/sme-fundamentals/sme-definition_en

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("Labour.csv")

# Size bands by the number of workers: fewer than 50, 50 to 249, 250 or more
df["size_band"] = pd.cut(df["labour"], bins=[0, 50, 250, np.inf],
                         labels=["Small", "Medium", "Large"],
                         right=False)

# The order of the size bands, from small to large
size_order = ["Small", "Medium", "Large"]

print(df.shape)
df.head()
```

The result must be `(569, 5)`. If you see an error instead, check the name of the file and that it sits in the same folder as the notebook.

How to work

- Every task names the type of chart you need. Find the code for this chart in the materials of the seminar (the demonstration notebook and the cheat sheet), copy it and change only the names of the columns and the texts of the titles.
- Every chart needs a title and axis labels with units.
- Every task ends with a question. Answer it in one or two sentences in a markdown cell under the chart. The delegation will read the sentences, not the code.

Task 1. How much does a worker cost?

Chart type: histogram, box plot, two charts in one figure (subplots).

1. Create a figure with two panels side by side: `fig, axes = plt.subplots(1, 2, figsize=(12, 5))`.
2. On the left panel draw a histogram of `wage` with 20 bins.
3. On the right panel draw a box plot of `wage`.
4. Give each panel a title and axis labels. The unit of `wage` is thousands of euro per worker.

Question: between which values does the middle half of the wages lie? Are there dots beyond the whiskers, and on which side are most of them?

Task 2. What kind of firms will the delegation meet?

Chart type: bar chart in the correct order.

1. Count the firms in each size band with `.value_counts()`.
2. Put the bars in the order Small, Medium, Large with `.reindex(size_order)`.
3. Draw the bar chart with `.plot(kind="bar")`, keep the labels horizontal and add a title and axis labels.

Question: which size band has the most firms? Would the bars be in the right order without `.reindex(size_order)`?

Task 3. Do larger firms pay more?

Chart type: box plots by group.

1. Draw box plots of `wage` by `size_band`: `size_band` on the horizontal axis, `wage` on the vertical axis.
2. Use `order=size_order` so that the boxes go from Small to Large.
3. Add a title and axis labels with units.

Question: compare the three medians (the lines inside the boxes). In which size band are wages highest? Is the difference between Small and Medium large or small?

Task 4. Do firms with more workers produce more?

Chart type: scatter plot, scatter plot coloured by a category (hue).

1. Draw a scatter plot with `labour` on the horizontal axis and `output` on the vertical axis.
2. Add a title and axis labels with units.
3. Draw the same scatter plot a second time, now with `hue="size_band"` and `hue_order=size_order`, so that the dots are coloured by size band.

Question: do firms with more workers produce more? Find the dot that lies far to the right of all the others: approximately how many workers does this firm have, and to which size band does it belong?

Task 5. What goes together with what?

Chart type: correlation heatmap.

1. Select the four numerical columns: `num_df = df[["capital", "labour", "output", "wage"]]`.
2. Compute the correlation matrix with `.corr()`.
3. Draw it as a heatmap with the numbers in the cells (`annot=True`, `fmt=".2f"`), the colour map `"coolwarm"` and the colour scale from -1 to 1.

Question: which pair of columns has the strongest correlation? How strong is the correlation between `wage` and `output`? Take a first look at question 4 of the delegation from seminar 2: do higher wages go together with higher output?

The deliverable

A notebook with five charts and five short answers. Read your answers once more as if you were the head of the delegation: each one must be understandable without the code.